Euclidean Geometry In Mathematical Olympiads

Problem 1.43 (JMO 2011/5)

Chapter 1 fgm fgmcon

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Mar 16, 2019, 6:25 PM • 2 #1

Points A, B, C, D, E lie on a circle ω and point P lies outside the circle. The given points are such that (i) lines PB and PD are tangent to ω, (ii) P, A, C are collinear, and (iii) $DE \parallel AC$. Prove that BE bisects AC.

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Solution

Let $M = BE \cap AC$ and O be center of ω, then $\measuredangle OBP = \measuredangle ODP$ and $\measuredangle PDB = \measuredangle DEB = \measuredangle PMB \implies PBMOD$ cyclic, so $OM \perp PC \implies AM = MC$, so we are done.

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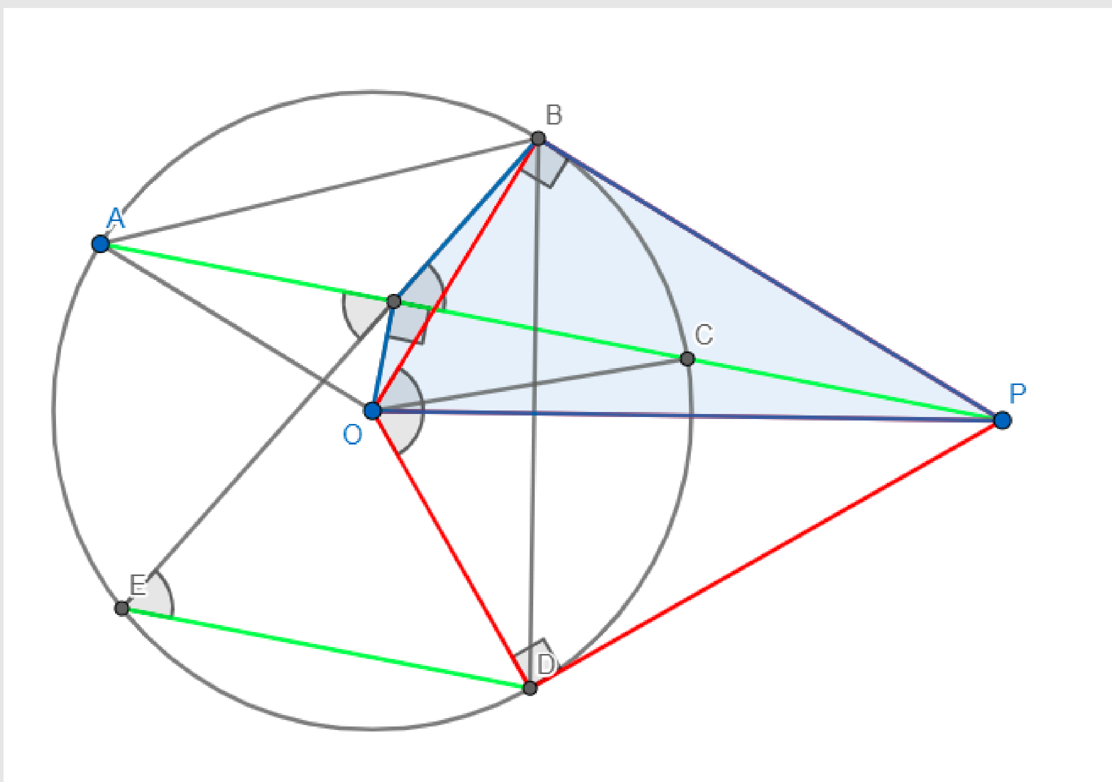
3001 posts

Mar 16, 2019, 6:40 PM • 2 #3

I could have sworn that there was already a thread for this, but whatever

Solution 1 (Angle Chasing)



Let M be the intersection of BE and AC, O be the center of ω, and $\angle POB = x$. It suffices to prove that $OMBP$ is cyclic, as that would imply that $\angle OMP = \angle OBP = 90$, which would then imply that M is the midpoint of AC, since a radius perpendicular to a chord bisects it. First, note that $BOPD$ is a cyclic quadrilateral because $\angle PBO = \angle PDO = 90$ due to PB and PD being tangents to ω and that $\triangle OBP \cong \triangle ODP$ by HL congruence ($PB = PD$ from equal tangents and OP is a shared side). Then, by inscribed angles, $\angle DEM = \dfrac{\angle DOB}{2} = x$. However, we also have $\angle DEM = \angle PMB = x$ since $BA || DE$, so $OMBP$ is cyclic and we're done. $\square$

since $PA||DE$, so $OMBP$ is cyclic and we're done. $\square$

Solution 2 (Symmedians)

First of all, AC is clearly the A symmedian w.r.t. $\triangle ADB$, so it suffices to show that $\angle ABM = \angle DBC$ since BD is also the B symmedian w.r.t. $\triangle ABC$. However, $\angle AMB = \angle 180 - \angle BED = \angle DCB$ and $\angle BAM = \angle BDC$, so we're done by AA similarity. $\square$

This post has been edited 1 time. Last edited by Kagebaka, May 25, 2019, 8:00 PM

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1443 posts

Aug 18, 2019, 2:06 PM • 3 #4

Solution 1 (Angle Chasing)

Solution 2 (Harmonics)

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